

Bayesian Decision Theory

Complete Notes with Concept Explanation, Formula Breakdown & Solved Example (Spam Detection)

1. What is it actually doing? (Intuition)

Suppose we want to build a **Spam Email Detector**. We have some training data:

Email	Class
"Win \$10,000 now"	Spam
"Meeting at 5 am"	Not Spam
"Claim your prize"	Spam
"Project report attached"	Not Spam

Now a **new email** arrives: *"Congratulations! You won \$50,000"*

Question: is it **Spam** or **Not Spam**? To answer this mathematically (not just guessing), we use **Bayesian Decision Theory**.

What we learn from the data:

- Words like *win, prize, money* → tend to appear in **Spam**
- Words like *meeting, project, report* → tend to appear in **Not Spam**

Bayesian Decision Theory gives us a formal way to combine this evidence with prior knowledge to make the best decision.

2. Core Idea

- Observed data** = feature vector x → we observe some evidence and want to choose the **best class**.
- Goal**: minimize classification error → i.e., choose the class with the **highest probability** given the observed data.

Simple Example

Let event **A** occur with probability 70%, and event **B** with 30%.
Since $P(A) = 0.7 > P(B) = 0.3$ → we select **Class A** (the more probable one).

3. The Bayes Formula

$$P(C_k \mid x) = \frac{P(x \mid C_k) \cdot P(C_k)}{P(x)}$$

Posterior — $P(C_k \mid x)$	[Most Important] Probability that class is C_k , given we observed data x
Prior — $P(C_k)$	Probability of class C_k <i>before</i> seeing any data
Likelihood — $P(x \mid C_k)$	Probability of observing feature x <i>given</i> the object belongs to class C_k
Evidence — $P(x)$	Overall probability of observing feature x (across all classes)

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$$

Evidence — Total Probability

$$P(x) = \sum_k P(x \mid C_k) \cdot P(C_k)$$

Note: Since $P(x)$ is the same for every class, it acts as a **normalizing constant** — when just comparing classes, it can be ignored, but for actual probability values it's needed so that all posteriors sum to 1.

4. Two-Category Classification & Decision Rule

For 2 classes C₁ and C₂:

If P(C₁ | x) > P(C₂ | x) → assign x to **C₁**

If P(C₁ | x) < P(C₂ | x) → assign x to **C₂**

Likelihood Ratio Form

Dividing both sides by common terms, the decision rule splits into two parts:

$$\frac{P(x | C_1)}{P(x | C_2)} > \frac{P(C_2)}{P(C_1)}$$

↑ Likelihood Ratio ↑ Threshold

5. Gaussian (Normal) Likelihood

In pattern recognition, feature data is usually assumed to follow a **Normal Distribution**. The probability density is:

$$P(x | C_k) = \frac{1}{\sqrt{(2\pi\sigma_k^2)}} \exp \left(-\frac{(x - \mu_k)^2}{2\sigma_k^2} \right)$$

μ (mu)	Mean
σ (sigma)	Standard deviation

Why each part is there:

- x − μ** → distance of x from the mean
- (x − μ)²** → makes it positive (distance can't be negative)
- divided by σ²** → normalizes according to spread of the data
- exp(·)** → converts this into a proper probability curve (bell shape)

6. Fully Solved Example — Spam Detection

Setup

Class 1: **Spam** | Class 2: **Not Spam (Ham)**

Total dataset = 100 emails

Step 1 — Prior Probability		
Category	Count	Prior Probability
Spam	40	P(Spam) = 40/100 = 0.40
Not Spam (Ham)	60	P(Ham) = 60/100 = 0.60

Step 2 — Likelihood (word "free" appears)			
Category	Total Emails	Contain word "free"	Likelihood
Spam	40	30	P(free Spam) = 30/40 = 0.75
Ham	60	10	P(free Ham) = 10/60 ≈ 0.17

Step 3 — Evidence (total probability of word "free")

$$\begin{aligned} P(\text{free}) &= P(\text{free}|\text{Spam}) \cdot P(\text{Spam}) + P(\text{free}|\text{Ham}) \cdot P(\text{Ham}) \\ &= (0.75 \times 0.40) + (0.17 \times 0.60) \\ &= 0.30 + 0.102 \\ &= \mathbf{0.402} \end{aligned}$$

Step 4 — Posterior Probability

For Spam:

$$P(\text{Spam} | \text{free}) = \frac{P(\text{free}|\text{Spam}) \times P(\text{Spam})}{P(\text{free})} = \frac{0.75 \times 0.40}{0.402} = \frac{0.30}{0.402} = \mathbf{0.746}$$

For Ham:

$$P(\text{Ham} | \text{free}) = \frac{0.17 \times 0.60}{0.402} = \frac{0.102}{0.402} = \mathbf{0.284}$$

Spam = 74.6% ✓ (Chosen)
Ham = 25.4%

Decision Rule → Take the class with the higher posterior probability

7. Bayes' Theorem — General Form

$$P(\text{Class} | \text{Data}) = \frac{P(\text{Data} | \text{Class}) \times P(\text{Class})}{P(\text{Data})}$$

Data by itself doesn't decide the class directly — instead, the class with the **highest posterior probability** is chosen.

8. Real-Life Applications

- Spam Email Detection
- Image Classification
- Fraud Detection
- Self-Driving Cars
- ...and many more pattern recognition tasks

Quick Recap (Hindi): Bayesian Decision Theory batati hai ki jab humein kuch data (evidence) dikhta hai, toh us data ke aadhar par kaunsi class (jaise Spam ya Not Spam) sabse zyada probable hai. Hum Prior (pehle se pata probability), Likelihood (data class mein kitna common hai), aur Evidence (overall probability) ko combine karke Posterior probability nikalte hain — aur jis class ka posterior sabse zyada hota hai, wahi final decision hota hai.